

Technical Document #502: Retrieval Augmented Generation - Comprehensive Overview

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Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Re-ranking models score retrieved documents for relevance to the query.

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.